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Modelling the impacts of environmental variation on the distribution of blue marlin, *Makaira nigricans*, in the Pacific Ocean

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Blue marlin are distributed throughout tropical and temperate waters in the Pacific Ocean. The preference of this species for particular habitats may affect its distribution and vulnerability to being caught. The relationships between the spatial pattern of blue marlin abundance and oceanographic conditions, which may be influenced by climate change, were examined using generalized additive models fitted to catch and effort data from longline fisheries. Distributions of blue marlin density, based on combining the probability of presence and abundance given presence, indicate that there is annual variation in the distribution of blue marlin and that the population apparently moved east during the 1997–1998 *El Niño*. The interannual variability in blue marlin distribution appears to be associated with *El Niño* events and is related to shifts in sea surface temperature and the deepening of the thermocline. Models of catch and effort that include oceanographic variables could be used, given predictions from climate models, to explore future changes in distribution, which could then be used to provide management advice related to time-area closures.

Keywords: climate change, generalized additive model, longline fisheries, oceanographic variables, spatial distribution.

Introduction

Blue marlin, *Makaira nigricans*, is a highly migratory cosmopolitan species distributed throughout tropical, subtropical, and temperate waters between 45°N and 45°S (Nakamura, 1985). They have been consolidated into a single species from two species in the Indo-Pacific and Atlantic Oceans, based on analyses of genetic divergence (Maguire *et al.*, 2006). Fishery catch rates, as well as molecular analyses (Graves and McDowell, 2003), suggest a single stock of blue marlin in the Pacific Ocean (Kleiber *et al.*, 2003). This assumption is also supported by the results of tagging experiments that have demonstrated that blue marlin migrate throughout the Pacific Ocean (Hinton, 2001).

Blue marlin are important commercial and recreational fish (Molony, 2005). They are caught primarily in pelagic longline fisheries targeting tuna *Thunnus* spp. and swordfish *Xiphias gladius*, although small catches are also taken in coastal fisheries that use gillnets, harpoons, and purse-seines (Hinton, 2001). Blue marlin exhibit dimorphic growth; female fish grow faster and to a larger size than males (Wilson *et al.*, 1991). The size-at-maturity of blue marlin also differs between the sexes (Shimose *et al.*, 2009; Sun *et al.*, 2009). The status of blue marlin in the Pacific Ocean remains unknown, with assessment results ranging from the stock being in a healthy condition (Hinton, 2001) to close to fully exploited (Kleiber *et al.*, 2003).

Tagging studies have elucidated the range and individual behaviour of blue marlin (Holland *et al.*, 1990; Block *et al.*, 1992). The preferences of this species for particular habitat-related factors may affect its distribution and vulnerability to being caught (Boyce *et al.*, 2008). Various oceanographic and biological variables have been hypothesized to affect the density and distribution of blue marlin, such as water temperature, ocean fronts, current speed, oxygen content, prey availability, and zooplankton abundance (Boyce *et al.*, 2008).

Water temperature influences habitat use by blue marlin (Block *et al.*, 1992), because they spend most of their time at the surface layer (temperatures >26°C), with occasional dives into deeper water. Seasonal latitudinal shifts in the distribution of blue marlin have been inferred from seasonal changes in sea surface temperature (SST; Goodyear, 2003; Su *et al.*, 2008). Water temperature also influences vertical movements. Specifically, blue marlin prefer the warm surface mixed layer above the thermocline and rarely range through the thermocline, although they can do so (Block *et al.*, 1992; Seki *et al.*, 2002).

Blue marlin might respond to the changes to oceanographic variables in an indirect way. For example, they feed primarily on prey found near the surface, such as small skipjack (*Katsuwonus pelamis*) tuna (Shimose *et al.*, 2006). Distribution and movement patterns of tuna-like species may be strongly correlated with environmental variation, *El Niño*–Southern Oscillation (ENSO) events, for instance (Lehodey *et al.*, 1997; Howell and Kobayashi, 2006). ENSO events could result in large-scale changes to various oceanographic features, such as an eastward shift in warm surface waters along the equatorial Pacific Ocean, a deepening mixed layer, elevated sea surface height, and lowered chlorophyll *a* concentration (CHL; Wilson and Adamec, 2001). Therefore, it might be anticipated that blue marlin react to changes in habitat characteristics and forage availability through the shifts in distribution.

Models of catch rates that incorporate relevant oceanographic variables associated with environmental variation could be used to infer possible responses in the distribution of a highly migratory species, such as blue marlin. These models can be used to explore future changes in the spatial pattern given adequate predictions from climate models. This study examines the functional relationships between the distribution of blue marlin and various oceanographic features that may be influenced by climate variation, and it uses these relationships to predict distribution. These functional relationships can be used to evaluate the impacts of climate change on the spatial pattern and vulnerability of blue marlin.

Material and methods

Catch and effort data for the Pacific longline fisheries grouped by $5^\circ \times 5^\circ$ grid cell (Figure 1) and by month (Figure 2) were obtained from the Oceanic Fisheries Programme of the Secretariat of the Pacific Community (http://www.spc.int/oceanfish/). Data for 1998–2004 were used for the analyses, because satellite-based CHL and mixed layer depth (MLD) are not available before 1998 and because the fisheries data after 2004 are restricted in the WCPFC convention area. [The fisheries data aggregated over nations have been held by WCPFC (Western and Central Pacific Fisheries Commission; http://www.wcpfc.int/) since 2004, but are restricted in the WCPFC convention area.] These data are aggregated over fleets, such as those of Japan, Taiwan, and the United States, because of confidentiality restrictions. This dataset contains information on time (year and month), location (latitude and longitude), fishing effort (in hooks), and catch (in number). Seasonal variation in the longline fisheries was examined by fitting a smooth spline (Hastie and Tibshirani, 1990) to monthly fishing effort (i.e. the number of hooks deployed) and nominal catch rates of blue marlin.

The oceanographic variables considered in this study [CHL, MLD, sea surface height anomaly (SSH), and SST] may be affected by environmental variation. The oceanographic data were averaged to a 5° grid to correspond to the spatial grid used for catch per unit effort (cpue), which was expressed as the number of fish caught by 5° grid cell divided by the total number of hooks deployed.

Monthly oceanographic data for 1998–2007 were sourced as follows: (a) CHL (1° resolution) from the SeaWiFS dataset (http://reason.gsfc.nasa.gov/); (b) MLD (10 min resolution) from the TOPS (http://www.science.oregonstate.edu/ocean.productivity/); (c) SSH (15 min resolution) from the AVISO (http://coastwatch.pfeg.noaa.gov/) dataset; (d) SST (1° resolution) from the Reynolds/NCEP provided by NASA/JPL PO-DAAC Pathfinder database (http://poet.jpl.nasa.gov/).

ENSO events strongly affect oceanographic features, especially in the equatorial Pacific Ocean (Wilson and Adamec, 2001). Therefore, contour maps of oceanographic conditions in the tropical Pacific Ocean (15°N–15°S) by year were created using monthly $5^\circ \times 5^\circ$ oceanographic data for 1998–2007. Medians (chosen to eliminate any extreme values) of those oceanographic data (over 1998–2007) were taken for each month and used to construct longitude–month isopleths to represent a year-invariant, but seasonal, marine environment.

Statistical models for spatial prediction

Statistical models were used to predict the spatial pattern of blue marlin density. The two most common approaches for doing this are generalized linear models (GLMs; Nelder and Wedderburn, 1972) and generalized additive models (GAMs; Hastie and Tibshirani, 1990). GAMs rather than GLMs were used in this study because GAMs can explain the variance of a response variable more effectively and flexibly, although it is more difficult to obtain confidence intervals and apply hypothesis tests using GAMs (Maunder and Punt, 2004). GAMs can deal with multiple non-linear relationships between the covariates and the response variable in a semi-parametric manner (Hastie and Tibshirani, 1990).

The "delta" method, based on a two-stage GAM (Ortiz and Arocha, 2004), was used to deal with the fact that some

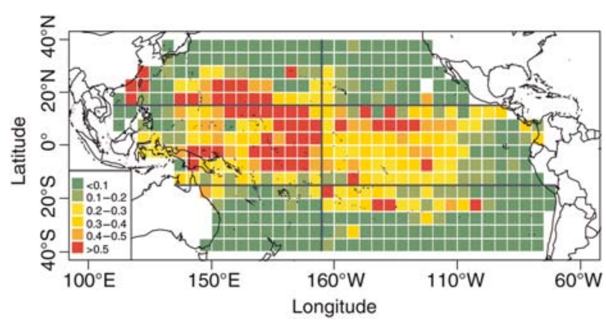

Figure 1. Distribution of nominal cpue (number of fish caught per 1000 hooks) for blue marlin in the Pacific longline fisheries (1998–2004). The straight lines separate the Pacific Ocean into six regions (see text for details).

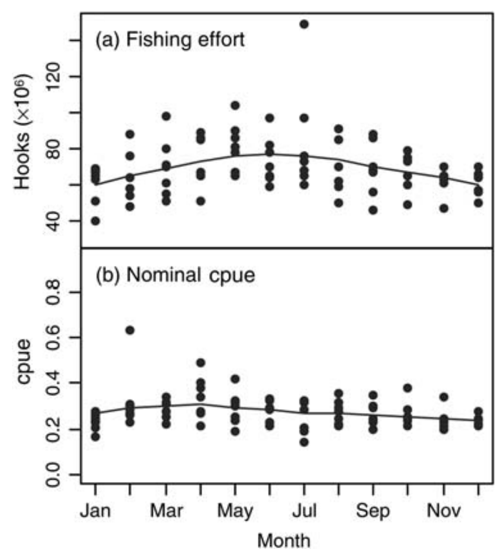

Figure 2. Monthly fishing effort (in hooks) and nominal cpue (number of fish caught per 1000 hooks) for blue marlin in the Pacific longline fisheries (1998–2004). The continuous lines are the results of fitting a smooth function with the tension factor $\lambda = 0.5$.

observations have no catch of blue marlin, but some effort. First, a GAM with a binomial response was used to analyse the data on the presence or the absence of blue marlin (presence/absence, P/A model), which determines the probability of catching blue marlin as a function of various covariates. Because blue marlin is a non-target species, the use of the P/A model could reduce the potential impact of changes in targeting practices when estimating abundance indices (Maunder and Punt, 2004). Several models were then used to model the catch rate given that it was positive (cpue model), based on the lognormal, gamma, and Poisson error models (Ortiz and Arocha, 2004). The dependent variable was taken to be the number of fish caught per record when using the Poisson error model. The P/A and cpue models can be written as:

$$P/A \sim s(\text{Year}) + s(\text{Month}) + s(\text{CHL}) + s(\text{MLD}) + s(\text{SSH}) + s(\text{SST}) + s(\text{Year, Latitude}) + s(\text{Year, Longitude}) + s(\text{Latitude, Longitude}) \quad \text{and} \tag{1}$$

$$\text{cpue} \sim s(\text{Year}) + s(\text{Month}) + s(\text{CHL}) + s(\text{MLD}) + s(\text{SSH}) + s(\text{SST}) + s(\text{Year, Latitude}) + s(\text{Year, Longitude}) + s(\text{Latitude, Longitude}), \tag{2}$$

where P/A is zero if no blue marlin were caught (one otherwise), cpue the catch rate of blue marlin if some blue marlin were caught, and s(X) denotes a spline smoother function of the covariate X or the interaction between two covariates. Year, latitude, and longitude were treated as interaction terms to account for conceivable interannual variability in the spatial distribution of blue marlin, probably driven by environmental variation.

The "best" set of the explanatory variables for the P/A and cpue models was selected in a forward stepwise manner. The predictors were considered to be appropriate in explaining the variance of catch rates if the residual deviance and Akaike's Information Criteria (AIC) decreased and the Chi-squared (χ^2) test was significant ($p < 0.01$) with each addition of a variable. All covariates were treated as continuous, and the effective degrees of freedom were estimated for each main factor. Diagnostic plots, i.e. the distribution of residuals and quantile–quantile (Q–Q) plots, were used to evaluate the model fits and select the most appropriate error distribution for the non-zero catch rates.

The selected P/A and cpue models were then used to predict the distribution of blue marlin abundance (relative density) over the Pacific Ocean, i.e. the P/A model was used to predict the probability of the presence of blue marlin and the cpue model was used to predict abundance given presence. Distributions of relative density, combining the probability of presence and abundance given presence, were predicted by month during 1998–2007 for every $5^\circ \times 5^\circ$ grid cell for which oceanographic data were available in the Pacific Ocean, by multiplying the predictions from the two models together.

Evaluating impacts of environmental variation

Two sets of oceanographic variable were used to examine how the distribution of blue marlin was influenced by *El Niño* events. The first ("original") was the month-varying oceanographic variables during 1998–2007 and the other ("contrast") was based on the medians over 1998–2007. The "original" oceanographic variables vary annually, whereas the "contrast" variables represent an annually static marine environment, with only seasonal variation in oceanographic variables. Impacts of environmental variation can be evaluated by comparing the predictions for the two sets of environmental variables.

The predicted relative densities by 5° grid cells were averaged by six regions, which separate the Pacific Ocean into tropical and subtropical waters at 15°N and 15°S and nearly equally into eastern and western Pacific Ocean at 165°W (Kleiber *et al.*, 2003; Figure 1). The impacts of environmental variation on the spatial patterns of blue marlin and their catch rates in longline fisheries were examined for each region. Relative changes were computed for each month during 1998–2007 as the difference between the region-averaged predictions based on original and contrast datasets, expressed as a fraction of the region-averaged predictions using contrast data.

The computational and presentational tasks were conducted using R version 2.11.0 (R Development Core Team, 2010).

Results

The total number of observations used in the analyses was 21 377, of which 15 824 indicate a catch of at least one blue marlin. The longline fisheries are distributed over much of the Pacific Ocean (Figure 1) and they operate continuously during the year, with some seasonal variation (Figure 2a). Monthly variation in nominal catch rates of blue marlin is quite minor when the data are averaged spatially (Figure 2b). The highest nominal catch rates of blue marlin are in the tropics (Figure 1).

There was interannual variability in the oceanographic variables during 1998–2007 (Figure 3; left panels). For example, warm surface waters and reduced chlorophyll *a* extended to the eastern tropical Pacific Ocean during the 1997–1998 *El Niño*, sea surface height was elevated, and the mixed-layer deepened. Seasonal changes in SST and MLD, but not in SSH and CHL, were evident when the data were aggregated over 1998–2007 (Figure 3; right panels).

The residuals from the lognormal distribution conform to the greatest extent with its assumptions according to Q–Q plots and because the residuals (in log-space) for the lognormal error distribution appear normal (results not presented). Therefore, remaining analyses of the positive catch rates are based on the lognormal distribution.

The model selection processes for the P/A and cpue models are summarized in Table 1, which lists the total deviance explained, the AIC, the percentage of the total deviance explained, and the *p*-values derived from a Chi-squared test between models that differ by one additional factor. The main effects and interaction terms are significant at $\alpha = 0.01$ in both models (Table 1). The total deviance explained by the cpue model for non-zero catches was 41.7%, whereas 33.7% of the deviance was explained by the P/A model.

Almost half of the explained deviance is associated with SST (22.4 and 16.2% for the two models, respectively) and omitting SST from the model resulted in the largest change in AIC. The oceanographic variable with the second highest contribution to the explained deviance in both models is MLD (4.7 and 5.6%). Spatial interannual variability that was not associated with any proposed oceanographic variables was expressed through the two-way interaction terms among year, latitude, and longitude, which altogether accounted for 10.7 and 9.2%, respectively, of the explained deviance in each model.

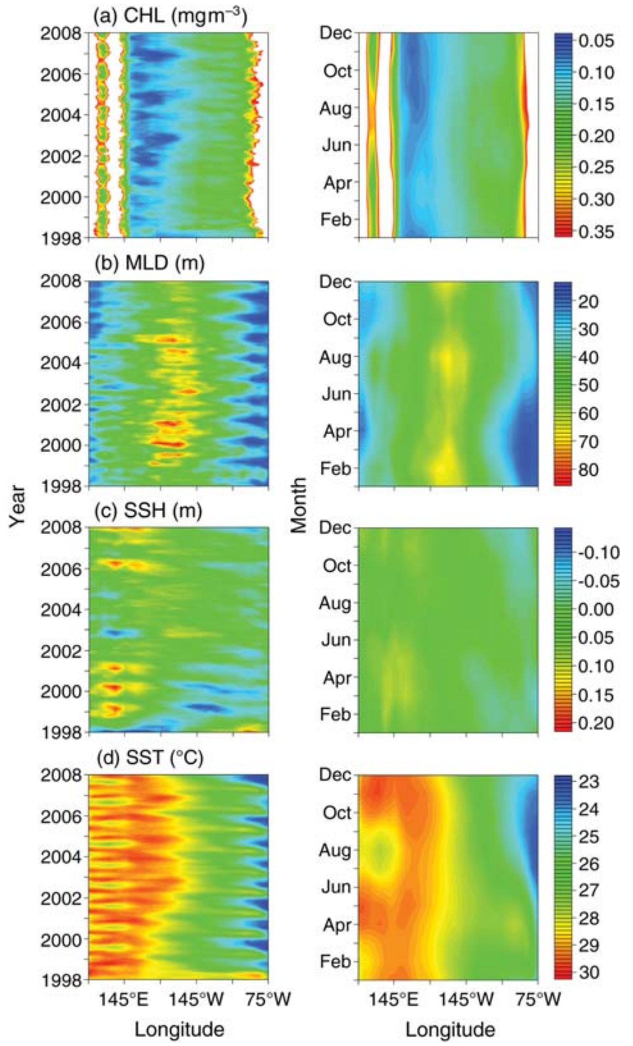

Figure 3. Isopleths of oceanographic data in the Pacific Ocean between 15°N and 15°S: (a) CHL, (b) MLD, (c) SSH, and (d) SST by year (left column) and month (right column).

The effects of the oceanographic factors are similar on the probability of the presence of blue marlin and its relative density given presence (Figure 4). However, the effect of SST differs slightly between P/A and cpue models and, for chlorophyll *a* levels $<0.1\ \mathrm{mg\ m^{-3}}$, the likelihood of blue marlin occurring increases with increasing CHL levels, but abundance decreases. SST was the main variable determining the distribution of blue marlin.

Predicted spatial distribution of blue marlin

The nominal catch rates and the predicted relative density of blue marlin by 5° grid cells in the Pacific Ocean are displayed in Figures 5 and 6 for April and October, respectively (because these months represent the extremes of the spatial distribution). The relative densities in Figures 5 and 6 were categorized into five equal-sized groups for ease of presentation and comparison.

The model predictions are much smoother than the nominal catch rates (Figures 5 and 6). Both the probability of catching a blue marlin and the catch rate given a non-zero catch vary spatially and temporally. The relative densities, based on combining the

Table 1. Analysis of deviance tables for the P/A and cpue models.

	Residual deviance	Percentage of total deviance explained (%)	AIC	$p(\chi^2)$
P/A model				
NULL	24 490		24 492	
+s(Year)	24 472	0.2	24 477	<0.01
+s(Month)	24 427	0.5	24 439	<0.01
+s(CHL)	23 967	5.6	23 985	<0.01
+s(MLD)	22 603	16.6	22 632	<0.01
+s(SSH)	22 459	1.7	22 494	<0.01
+s(SST)	18 494	48.1	18 532	<0.01
+s(Year, Latitude)	17 671	10.0	17 746	<0.01
+s(Year, Longitude)	16 888	9.5	17 001	<0.01
+s(Latitude, Longitude)	16 245	7.8	16 388	<0.01
Total deviance explained	33.7%			
cpue model				
NULL	27 595		53 711	
+s(Year)	27 378	1.9	53 593	<0.01
+s(Month)	27 318	0.5	53 563	<0.01
+s(CHL)	26 745	5.0	53 236	<0.01
+s(MLD)	25 446	11.3	52 456	<0.01
+s(SSH)	25 218	2.0	52 321	<0.01
+s(SST)	19 025	53.8	47 868	<0.01
+s(Year, Latitude)	17 929	9.5	46 967	<0.01
+s(Year, Longitude)	17 295	5.5	46 435	<0.01
+s(Latitude, Longitude)	16 076	10.6	45 297	<0.01
Total deviance explained	41.7%			

probability of catching a blue marlin and the catch rate given that the catch is non-zero, reveal a notable increase in both quantities in equatorial waters, particularly between 20°N and 20°S. Moreover, there is seasonal variation in the areas of highest relative density (see the dark grey grids in Figures 5 and 6), which suggests that blue marlin migrate north during April–October and south thereafter. This latitudinal movement appears to be associated with the seasonal shifts in oceanographic conditions, especially SST (Figures 3 and 4).

The predicted year-invariant, but seasonally varying, distributions of blue marlin in the Pacific Ocean are illustrated in Figures 5d and 6d. The predicted relative densities indicate that there is interannual variability in the distribution of blue marlin related to environmental variation. The model predictions for April during 2001–2004 are similar to those that ignore interannual variation in oceanographic conditions, which might be a consequence of a reduced *El Niño* effect in those years. The blue marlin population was located more to the east during *El Niño* events (April 1998 and 2007; Figure 5e and h).

Impacts of environmental variation

According to our model, the impacts of environmental variation on the spatial distribution of blue marlin and its vulnerability to fisheries are quite minor for the southern Pacific Ocean (dashed lines in Figure 7), because the oceanographic variables do not

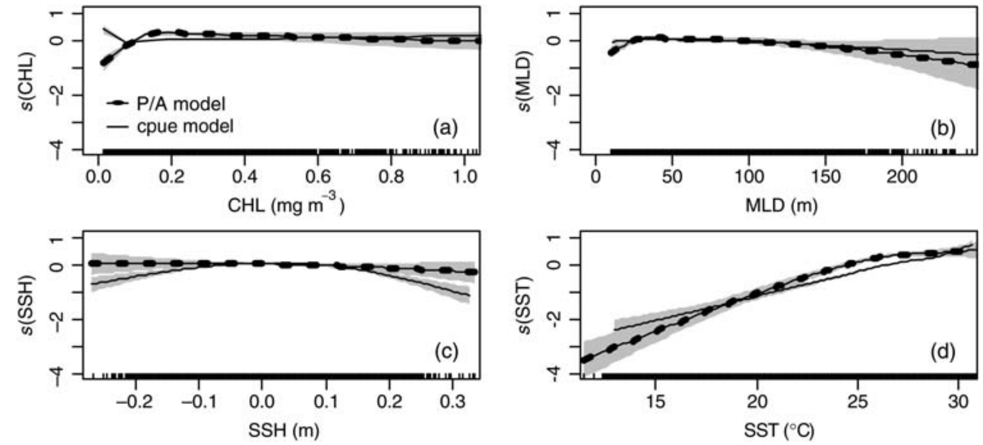

Figure 4. The effects of oceanographic variables inferred from the GAMs on the probability of presence (P/A model) and catch rates given presence (cpue model): (a) CHL, (b) MLD, (c) SSH, and (d) SST. Shaded areas indicate 95% confidence intervals.

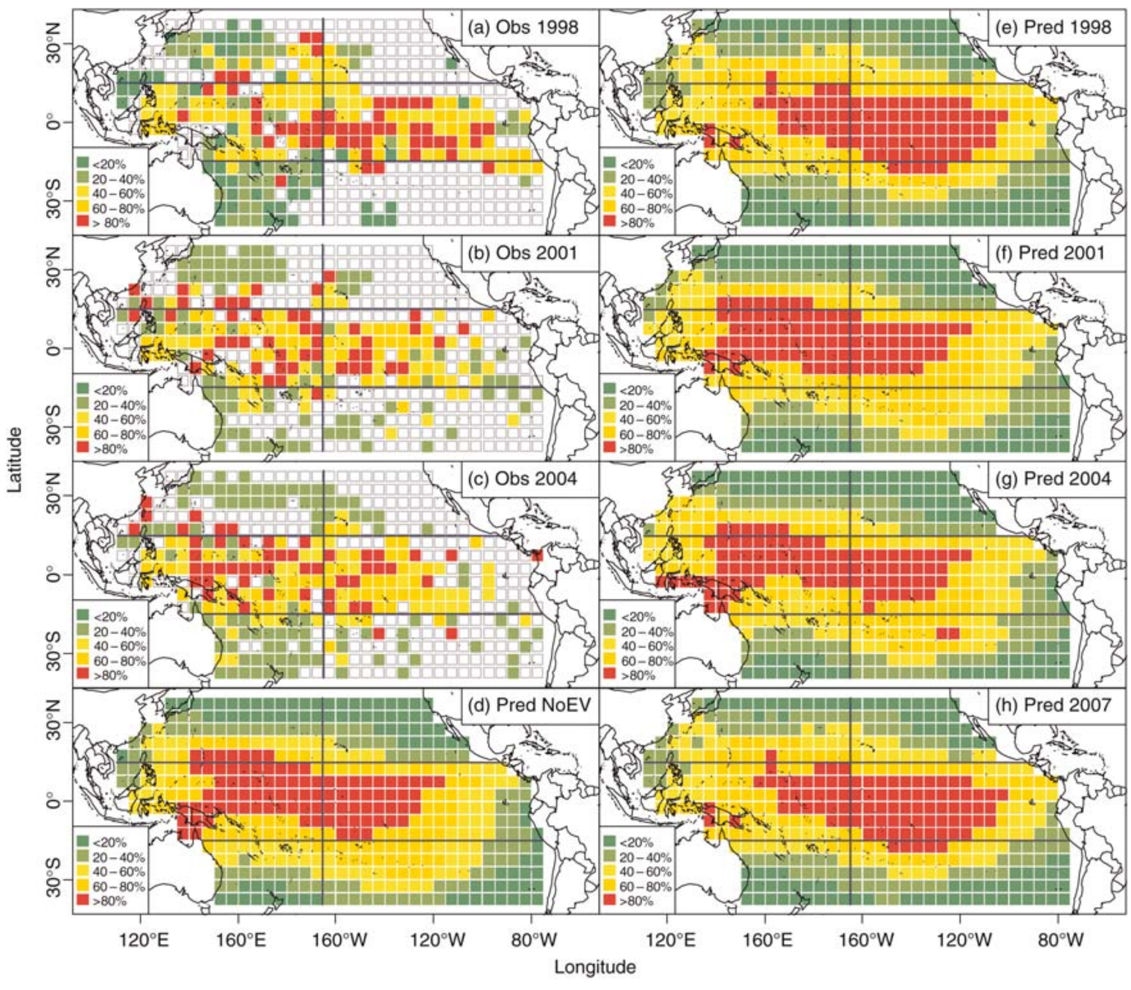

Figure 5. Relative density of Pacific blue marlin during April: (a)–(c) observed (nominal) distribution, (d) predicted values using GAMs based on median oceanographic data taken over 1998–2007, and (e)–(h) model predictions based on the original oceanographic data. The horizontal and vertical lines separate the Pacific Ocean into six regions (see text for details).

change substantially among years. However, substantial changes in oceanographic conditions resulted in decreased relative density of blue marlin in the northwest Pacific Ocean during 1999–2003 (dotted lines in the upper panel of Figure 7). Moreover, the predicted relative density of blue marlin in the eastern Pacific Ocean was higher during the 1997–1998 *El Niño* than the

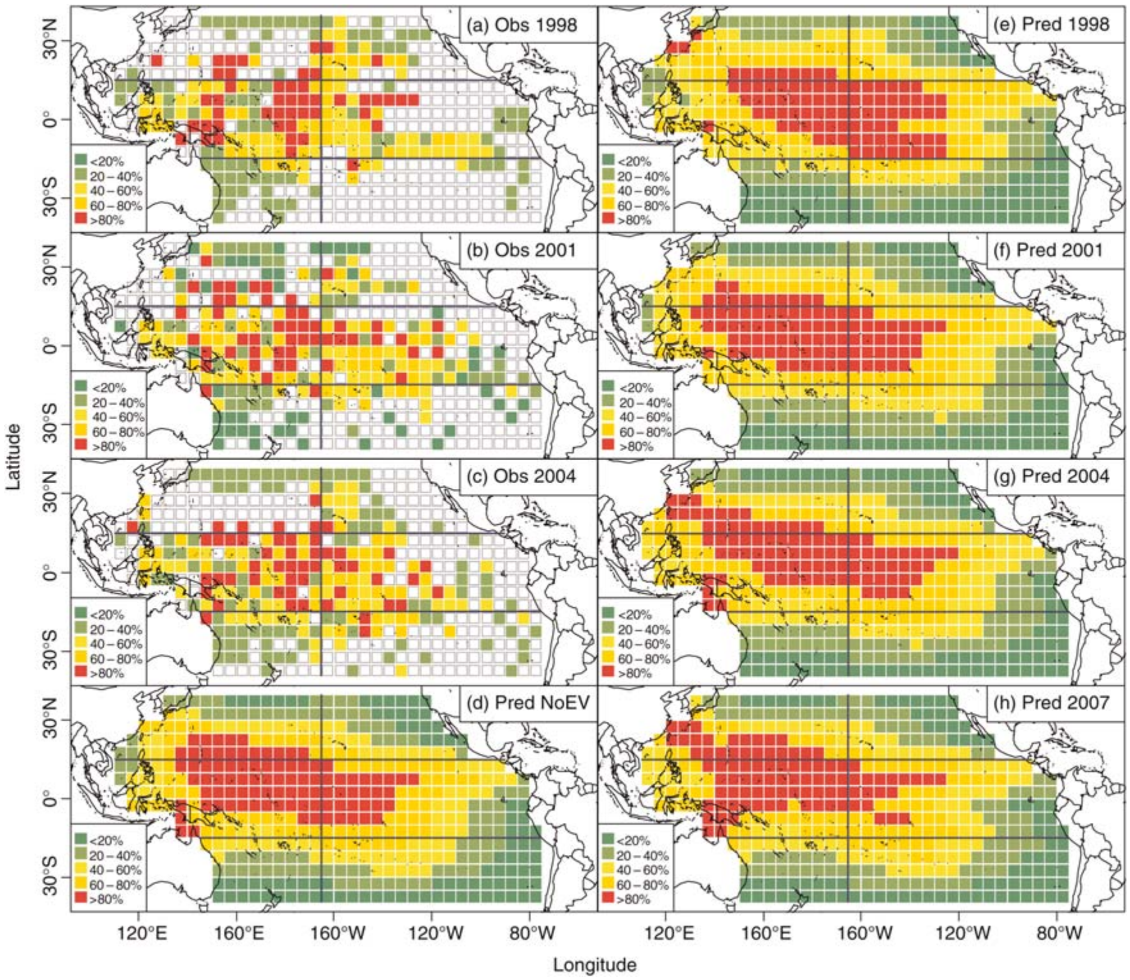

Figure 6. As for Figure 5, but for October.

predictions based on annually static environment (lower panel in Figure 7), which suggests an eastward movement of the blue marlin population into a preferred environment during those years. The spatial patterns of blue marlin have apparently been more normal since 2005, following the reduced ENSO effect (Figure 7).

Discussion

Although the fishery and oceanographic data used in this study start in 1998, this dataset covers a full ENSO cycle of 3–6 years, during which an *El Niño*/*La Niña* (from late 1997 to 1999) and normal conditions served (Wilson and Adamec, 2001). The impacts associated with ENSO events could be evaluated because the dataset for blue marlin includes data from an *El Niño* and the subsequent *La Niña*, as well as normal conditions. The catch and effort dataset includes data for several of the main longline fisheries in the Pacific Ocean and covers the distribution of this population. This dataset therefore provides a new way to explore potential impacts of environmental variability on changes in distribution for a cosmopolitan species such as blue marlin. However, it is likely that the use of finer-scale catch and effort data would increase the proportion of the deviance explained, although the explained deviance in this study is also typical or higher than what can be achieved when standardizing catch and effort data (AEP, pers. obs.). In addition, information on operational targeting practices, individual vessel data, and fishing strategies, along with other environmental variables that have been demonstrated to be related to blue marlin dynamics (e.g. dissolved oxygen concentration; Prince and Goodyear, 2006) had they been available, could be have been included in the modelling and should increase further the explained deviance.

The oceanographic variables on which this study was based are key predictors for pelagic species habitat (Polovina *et al.*, 2001). Climate variation and climate change could result in large-scale interannual variability and trends in oceanographic processes. For example, SSTs exceed normal levels, with an elevated sea level, lower CHLs, and deepening thermocline in the eastern tropical Pacific Ocean during *El Niño* years (Wilson and Adamec, 2001). Those variables have been demonstrated through tagging experiments (Holland *et al.*, 1990; Block *et al.*, 1992) and analyses of fishery catch rates (Su *et al.*, 2008) to affect the dynamics of blue marlin. Some oceanographic features may be related to factors that limit physiological processes directly or factors that govern forage abundance indirectly; they therefore play an important role in determining the distribution of blue marlin (Seki *et al.*, 2002; Su *et al.*, 2008). The oceanographic covariates considered in this

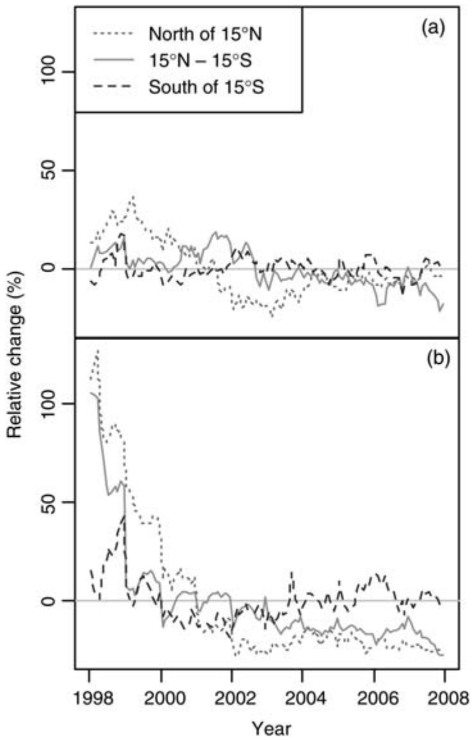

Figure 7. Relative changes in region-averaged model-predicted relative densities of blue marlin between predictions based on original variables and those based on the median data, for regions in the (a) western and (b) eastern Pacific Ocean.

study are strongly correlated with ENSO events (Wilson and Adamec, 2001).

The effect of environmental variation on the distribution of blue marlin and the fisheries that utilize the stock could be evaluated, because the statistical model included relevant oceanographic variables. Predicting the distribution of a species is of primary importance in ecology, conservation biology, and management of fisheries resources subject to climate change (Austin, 2007). Various modelling approaches, such as GLMs, GAMs, artificial neural networks (ANNs), and classification and regression tree analysis (CART—also referred to as classification tree analyses or decision trees; Thuiller, 2003), have been used to relate the patterns of species presence/absence and abundance to habitat-related variables.

Although GLMs are the most commonly used method for predicting species distribution, the inability to account for complex non-linear functions of covariates has stimulated increasing use of GAMs. ANNs have been used more frequently recently, but it is difficult to identify causal relationships and the dominant covariates from the network structure. The CART can deal with non-linear relationships among variables and can describe the hierarchical interactions among species (Thuiller, 2003). However, the CART must create discrete relationships, although continuous functions may be more realistic in some instances. Therefore, these approaches would not be ideal for producing spatial maps of relative density, such as those in Figures 5 and 6.

Temporal variation in spatial distribution

Blue marlin have the most tropical distribution of all billfish (Molony, 2005) and are the most common billfish in the catches of longline fisheries in the tropical Pacific Ocean (Hinton, 2001). The distributions of blue marlin density vary seasonally and annually (Table 1; Figures 5 and 6), because of interannual variability in oceanic conditions. Specifically, although the bulk of the population is found in warm tropical waters, blue marlin moved east during *El Niño* years, as evidenced by higher catch rates, but not effort, in the eastern area (results not presented).

As a pelagic surface-dwelling species, blue marlin most likely extend their geographic and vertical ranges along the eastward extent of warm surface waters and the deepening of the thermocline during ENSO events, and may become much more catchable to longline sets then (Ward and Myers, 2005). This assumption is confirmed by the eastward shift of the blue marlin population in tropical waters (Figures 5 and 6), in concert with the eastward expansion of warm surface waters and the deepening mixed layer during the *El Niño* period, which results in a substantial increase in catch rates.

Lehodey *et al.* (1997) found an apparent shift of the skipjack population towards the eastern equatorial Pacific Ocean that could be related to displacements of the warm pool and possibly the zonal transport of planktonic communities during ENSO events. This could be another plausible biological explanation, in addition to water temperature, for the eastward movement of blue marlin along the equator during the 1997–1998 *El Niño*. Blue marlin are generalist predators, but feed primarily on prey such as skipjack and small tuna in surface warm waters (Abitia-Cardenas *et al.*, 1999; Shimose *et al.*, 2006), and they most probably migrate following abundant forage. It is likely that blue marlin respond synoptically to environmental variation through a variety of oceanographic factors. Statistical models incorporating biophysical variables (physical, e.g. SSTs, and biological, e.g. chlorophyll *a*) might provide insights into the distribution of blue marlin; they therefore could be used to infer possible impacts and responses to climate change for such a highly migratory species.

Implications for management

Effective strategies for managing marine ecosystems will have to account for factors that influence the spatial distribution of the species. Goodyear (1999) suggested that time–area closures may be the best approach to manage the fisheries and protect species such as billfish that have pelagic larvae and highly migratory adults, because controlling fishing effort across a large geographic range seems impossible (Hobday and Hartmann, 2006). Jensen *et al.* (2010) found that temporary local closures could substantially increase the abundance of striped marlin in the eastern Pacific Ocean. In principle, the placement of reserves for migratory species such as blue marlin should be based on habitat characteristics, oceanographic conditions, migration patterns, and prey distributions to protect spawning and nursery grounds.

Environmental variation associated with regional or global phenomena, such as *El Niño* events, will result in large-scale shifts in distribution (as demonstrated in the current study). Therefore, modelling the distribution of highly migratory species should account for the interannual variability in oceanic condition. The relationships between oceanographic features and blue marlin abundance derived in this study could be used to predict the spatial distribution of this species and hence provide a basis to plan the closed areas. A large number of general circulation models (GCMs) have been developed to investigate the potential impact of increasing greenhouse gas concentrations on the climate system (IPCC, 2007). Some of the oceanographic variables derived from the current GCMs could be useful for predicting the future distribution of fish species. For example, Hare *et al*. (2010) used GCM-simulated air temperature as a proxy for water temperature to forecast the dynamics of Atlantic croaker (*Micropogonias undulatus*), Hobday (2010) examined the future distribution of large pelagic fish off Australia using surface ocean temperatures obtained from various GCMs, and Lehodey *et al*. (2010) modelled the impact of climate change for the Pacific bigeye tuna (*Thunnus obesus*) population using SST, primary production, and dissolved oxygen concentration predicted by a climate model coupled with an oceanic biogeochemical model.

Given appropriate predictions from GCMs, models of catch and effort that include oceanographic covariates, such as those developed in this paper, could be used to explore future changes in distribution for a highly migratory species, from which regional fisheries management organizations (IATTC and WCPFC in the Pacific Ocean) could develop time-area restrictions. Care should be taken when making such predictions to capture adequately both the uncertainty associated with the predictions from the climate models and that associated with the relationship between oceanographic variables and the distribution of the species of interest, as well as residual variation (e.g. the model in Table 1 includes year × latitude and year × longitude interactions, which suggests that there are factors other than oceanographic variables that determine spatial distribution).

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